

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0996

Perryman 138 kV
August 22, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0996. The Interconnection Customer proposes a 161.5 MW solar photovoltaic facility interconnecting at the Perryman 138 kV substation in the Maryland area. The request was received on May 14, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 381 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,759,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Windy Edge Sta-N Staunton 138 kV Line
Contingency: Fullerton Substation-Tanguisson 138 kV
Loading: 207.3% (Rating: 177 MVA)
DFAX: 5.9%
Pre-project loading: 202.0%
2. Facility: Fullerton-Tanguisson Substation 138 kV
Contingency: North Staunton-Windy Edge Sta 138 kV
Loading: 161.5% (Rating: 250 MVA)
DFAX: 6.0%
Pre-project loading: 157.6%
3. Facility: Notch Cliff-Windy Edge Sta 138 kV
Contingency: Hazelwood-Cloverleaf 138 kV
Loading: 159.5% (Rating: 265 MVA)
DFAX: 8.8%
Pre-project loading: 154.2%
4. Facility: Barling-Claremont 138 kV
Contingency: Windy Edge Sta-Notch Cliff 138 kV
Loading: 157.5% (Rating: 232 MVA)
DFAX: 6.4%
Pre-project loading: 153.0%
5. Facility: Joppatowne Substation-Raphael Road 138 kV Circuit 1
Contingency: Joppatowne-Raphael Rd Substation 138 kV Line 2
Loading: 153.8% (Rating: 277 MVA)
DFAX: 5.8%
Pre-project loading: 150.4%

6. Facility: Notch Cliff Substation-Caney Creek 138 kV L
Contingency: Hazelwood-Cloverleaf 138 kV
Loading: 152.5% (Rating: 236 MVA)
DFAX: 8.8%
Pre-project loading: 146.5%
7. Facility: Tanguisson-Raphael Road 138 kV Ckt 2
Contingency: Tanguisson-Raphael Rd 138 kV Circuit 1
Loading: 152.0% (Rating: 278 MVA)
DFAX: 6.0%
Pre-project loading: 148.5%
8. Facility: Tanguisson-N Staunton 138 kV L
Contingency: Fullerton Substation-Tanguisson 138 kV
Loading: 148.5% (Rating: 242 MVA)
DFAX: 5.9%
Pre-project loading: 144.6%
9. Facility: Tanguisson-Raphael Rd 138 kV Line 1
Contingency: Tanguisson-Raphael Rd 138 kV Ckt 2
Loading: 147.7% (Rating: 286 MVA)
DFAX: 6.0%
Pre-project loading: 144.3%
10. Facility: Raphael Rd-Joppatowne 138 kV #2
Contingency: Joppatowne-Raphael Road Substation 138 kV
Loading: 136.5% (Rating: 313 MVA)
DFAX: 5.8%
Pre-project loading: 133.4%
11. Facility: Fullerton-Windy Edge Sta 138 kV
Contingency: Windy Edge Sta-N Staunton 138 kV
Loading: 100.1% (Rating: 205 MVA)
DFAX: 6.9%
Pre-project loading: 94.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Windy Edge Sta-N Staunton Substation 138 kV	Rebuild line with higher-capacity conductor	\$880,000	12% (\$102,000)
Fullerton-Tanguisson Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,017,000	13% (\$133,000)
Notch Cliff-Windy Edge Sta 138 kV	Rebuild line with higher-capacity conductor	\$3,070,000	20% (\$627,000)
Barling-Claremont 138 kV	Rebuild line with higher-capacity conductor	\$4,083,000	46% (\$1,874,000)
Joppatowne Substation-Raphael Road 138 kV Circuit 1	Rebuild line with higher-capacity conductor	\$1,867,000	16% (\$298,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Notch Cliff-Caney Creek 138 kV line	Rebuild line with higher-capacity conductor	\$2,064,000	20% (\$421,000)
Raphael Rd-Tanguisson 138 kV	Rebuild line with higher-capacity conductor	\$6,134,000	16% (\$980,000)
Tanguisson-N Staunton Substation 138 kV line	Rebuild line with higher-capacity conductor	\$1,197,000	15% (\$176,000)
Tanguisson-Raphael Rd 138 kV Line 1	Rebuild line with higher-capacity conductor	\$4,143,000	16% (\$662,000)
Joppaowne-Raphael Rd 138 kV Line 2	Rebuild line with higher-capacity conductor	\$2,049,000	16% (\$327,000)
Fullerton-Windy Edge Sta 138 kV	Reconductor line	\$582,000	27% (\$159,000)
	Total allocated to project		\$5,759,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.